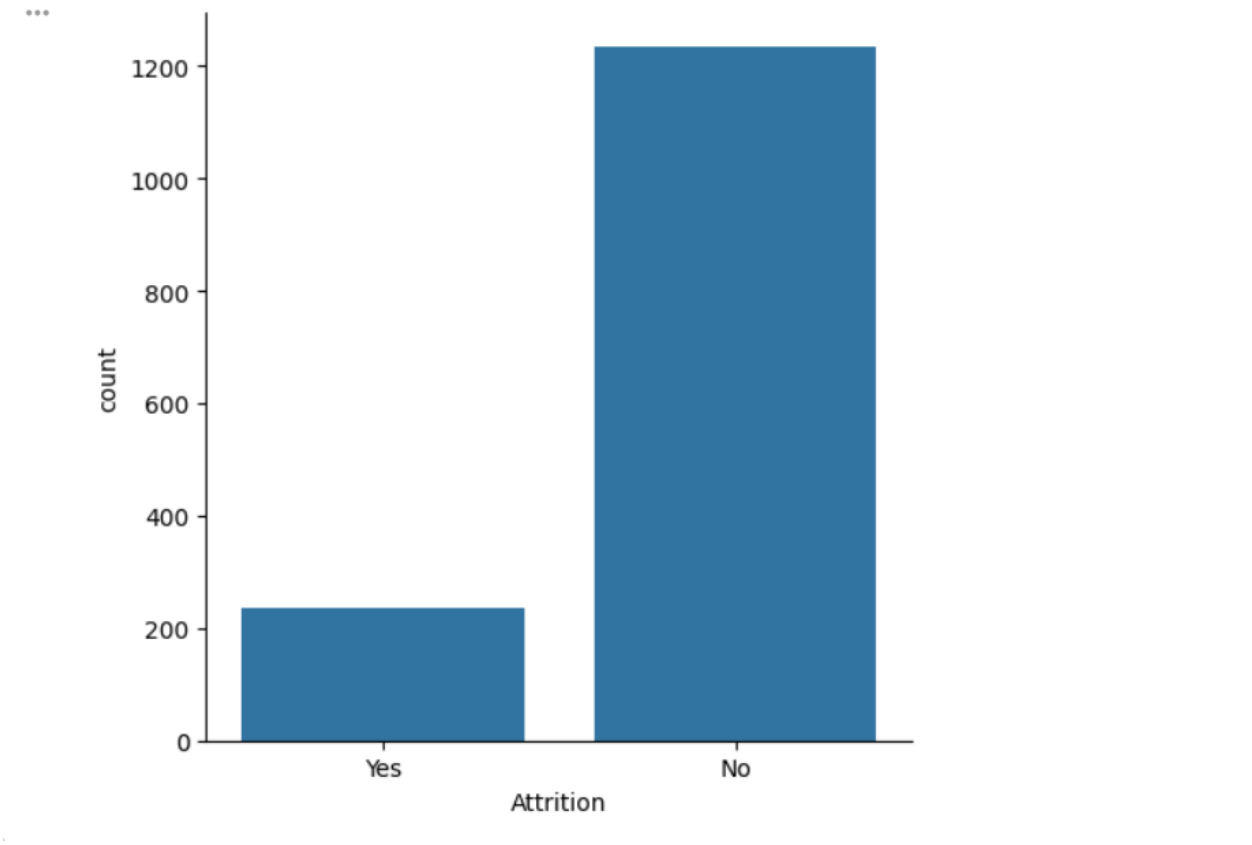
HR_Dataset_Analysis

Q1.What is the overall Attrition rate ??

ANS: 16.1% of employees left the company 237 out of 1470

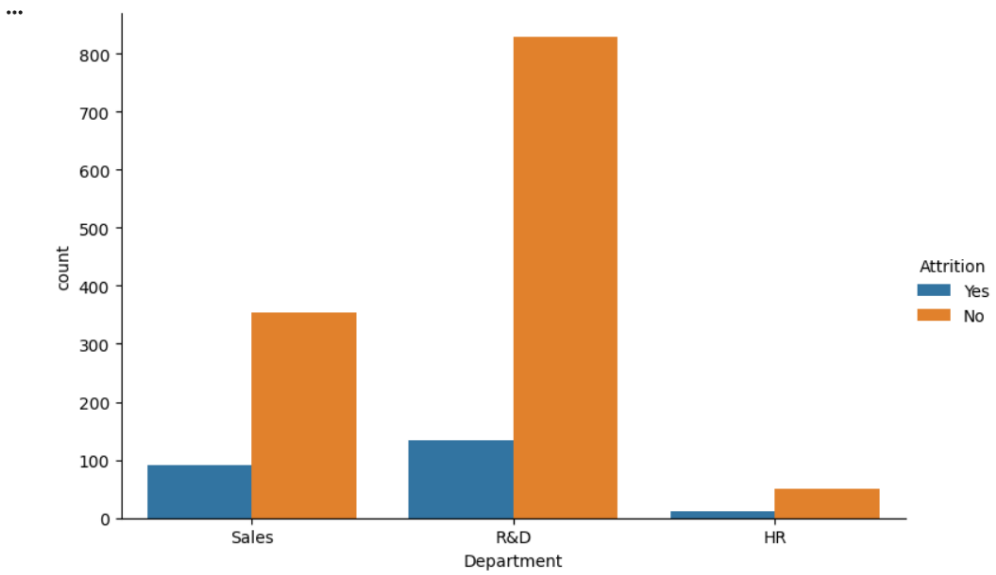
While 83.9% are current employees



Q2.Which department and job role have the highest attrition rate?

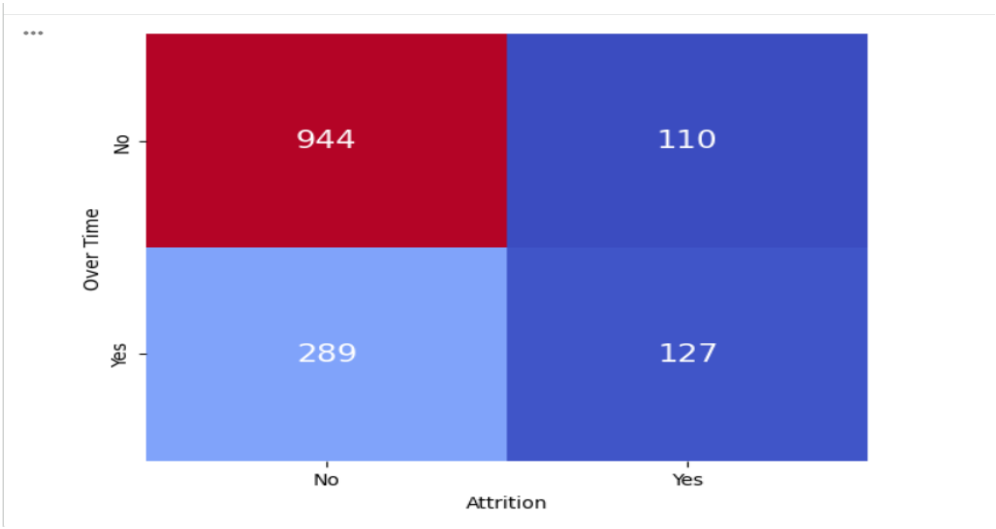
Ans: The Sales department has the highest attrition rate (20.6%) and sales Representative is the highest – attrition role (39.8%)

HR_Dataset_Analysis



Q3: How does OverTime relate to Attrition?

Answer: Employees who work OverTime leave at a rate of 30.5%, versus only 10.4% for those who don't



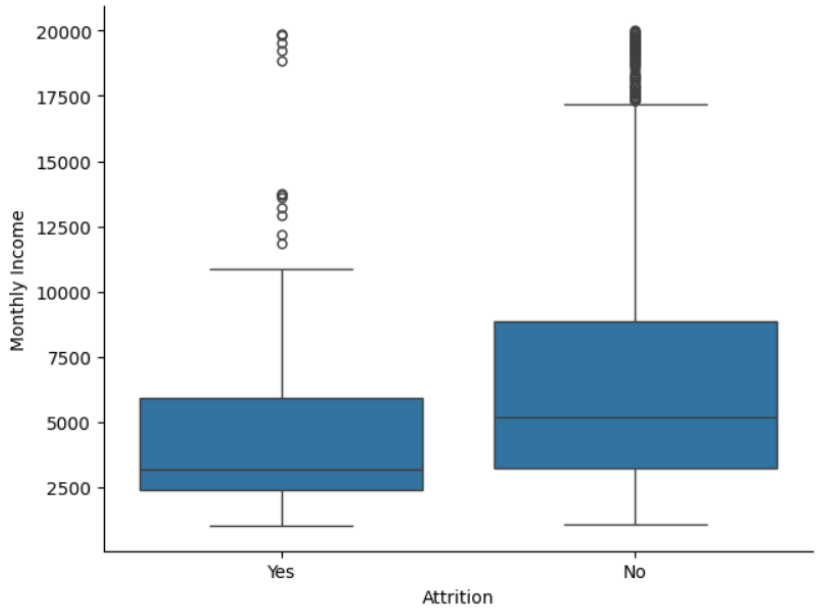
Q4: Does Monthly Income differ between employees who stayed vs left?

Answer: Average Monthly Income for employees who left is 4.787$ versus 6.833$ for those who stayed – lower pay is associated with higher attrition

HR_Dataset_Analysis

```
sns.catplot( data = df, x='Attrition', y='Monthly Income', kind='box', aspect=1.3 )

plt.show()
```

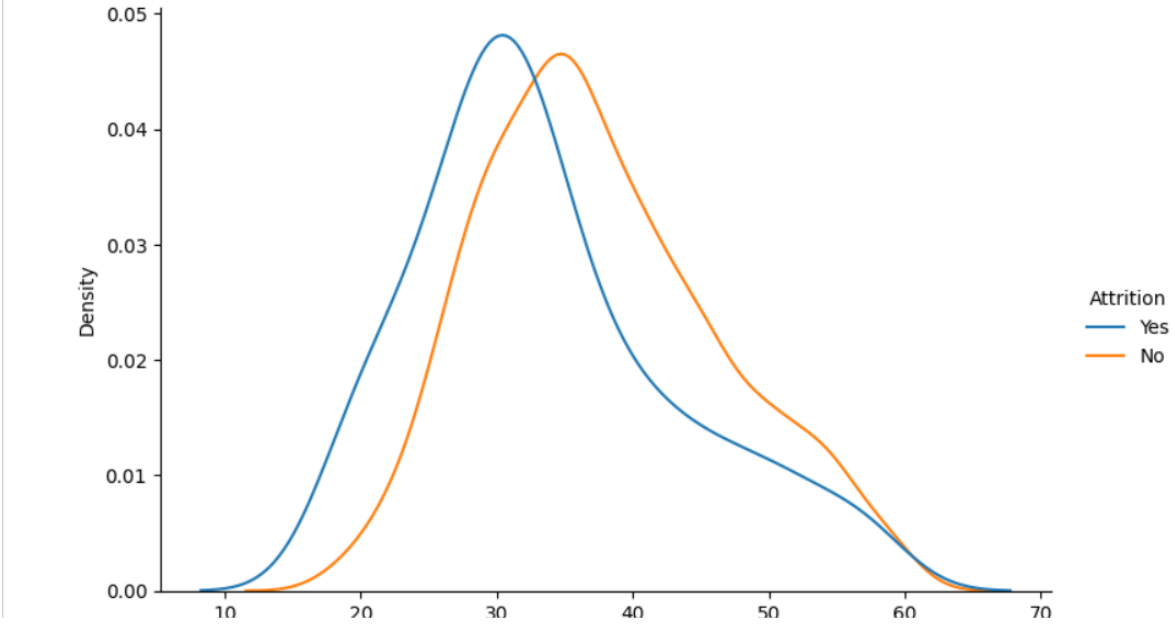


Q5: IS Age related to Attrition ?

ANS : Employees who left are younger on average (33.6 years) than those who stayed (37.6 years)

```
sns.displot( data = df, x='Age', kind='kde', hue='Attrition', common_norm=False, aspect=1.5 )

plt.show()
```



HR_Dataset_Analysis

Q6: Which numeric variables correlate most strongly with Attrition ?

ANS: All correlations are weak (below 0.2) , with the strongest being total Working years , job level , and monthly income (all negative)

